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Batch: A1 RollNo: 18

```
import time
import pandas as pd
from sklearn.datasets import fetch_california_housing
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
```

```
data = fetch_california_housing()
print(data.keys())
```

```
dict_keys(['data', 'target', 'frame', 'target_names', 'feature_names', 'DESCR'])
```

```
x = pd.DataFrame(data.data,columns = data.feature_names)
y = data.target
print(x.head())
print("samples:", len(x))
print(y[:10])
```

```
   MedInc  HouseAge  AveRooms  ...  AveOccup  Latitude  Longitude
0  8.3252      41.0  6.984127  ...  2.555556     37.88    -122.23
1  8.3014      21.0  6.238137  ...  2.109842     37.86    -122.22
2  7.2574      52.0  8.288136  ...  2.802260     37.85    -122.24
3  5.6431      52.0  5.817352  ...  2.547945     37.85    -122.25
4  3.8462      52.0  6.281853  ...  2.181467     37.85    -122.25

[5 rows x 8 columns]
samples: 20640
[4.526 3.585 3.521 3.413 3.422 2.697 2.992 2.414 2.267 2.611]
```

```
x_train,x_test,y_train,y_test = train_test_split(x, y, test_size = 0.2, random_state= 42)
```

```
tree = DecisionTreeRegressor()
start = time.perf_counter()
tree.fit(x_train, y_train)
tree_time = time.perf_counter() - start
pred_tree_test = tree.predict(x_test)
err_tree_test = mean_absolute_error(y_test, pred_tree_test)
pred_tree_train = tree.predict(x_train)
err_tree_train = mean_absolute_error(y_train, pred_tree_train)
print("Decision Tree Train MAE:" , err_tree_train)
print("Decision Tree Test MAE:", err_tree_test)
print("Decision Tree Time:", tree_time)
```

```
Decision Tree Train MAE: 4.217126217568424e-17
Decision Tree Test MAE: 0.45369783430232563
Decision Tree Time: 0.3055197629998929
```

```
rf = RandomForestRegressor()
start = time.perf_counter()
rf.fit(x_train, y_train)
rf_time = time.perf_counter() - start
```

```
pred_rf_test = rf.predict(x_test)
err_rf_test = mean_absolute_error(y_test, pred_rf_test)
pred_rf_train = rf.predict(x_train)
err_rf_train = mean_absolute_error(y_train, pred_rf_train)
print("Random Forest Train MAE:" , err_rf_train)
print("Random Forest Test MAE:", err_rf_test)
print("Random Forest Time:", rf_time)
```

```
Random Forest Train MAE: 0.1221313927143899
Random Forest Test MAE: 0.3262400572189924
Random Forest Time: 21.99320275199989
```

```
gb = GradientBoostingRegressor(n_estimators=300, learning_rate=0.1, max_depth=5)
start = time.perf_counter()
gb.fit(x_train, y_train)
gb_time = time.perf_counter() - start
pred_gb_test = gb.predict(x_test)
err_gb_test = mean_absolute_error(y_test, pred_gb_test)
pred_gb_train = tree.predict(x_train)
err_gb_train = mean_absolute_error(y_train, pred_gb_train)
print("Gradient Boosting Train MAE:" , err_gb_train)
print("Gradient Boosting Test MAE:", err_gb_test)
print("Gradient Boosting Time:", gb_time)
```

```
Gradient Boosting Train MAE: 4.217126217568424e-17
Gradient Boosting Test MAE: 0.30696441901489335
Gradient Boosting Time: 29.072496930999932
```